



Hybrid Perceptual World Models: World-Time Compute for Interactive Reasoning (ARC-AGI-3)

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ABSTRACT

ARC-AGI-3 is the ARC Prize Foundation’s interactive-reasoning benchmark: an agent dropped into an unfamiliar grid game must discover the rules *by acting* and reach goals over many steps; frontier agents score near zero. We study what a verified-world-model approach can and cannot do here, and our central contribution is a **diagnosis**: the bottleneck is not modeling but *goal discovery*, and specifically ARC-AGI-3 wins are **goal-as-procedure**, not goal-as-state.

(1) *Modeling is largely solved.* All 25 public games are replay-deterministic (1.00), so their dynamics are exactly code-expressible; a frontier synthesizer (Claude) writes verified `predict(frame,action)` code reaching 49% mean held-out exact-match on the real-dynamics games (vs. 12% for a local model), with 3 near-perfect ($\geq 90\%$) models. (2) *The wall.* Despite these models, **four principled goal-discovery methods**—atomic objectives, frontier-LLM hypotheses, Bayesian sub-world planning, and typed navigation—solve *zero* of the hard games, several *even on a fidelity-1.0 model*: a goal scored over a single *state* cannot express an ordered *procedure*, which is why undirected search occasionally stumbles into a win while goal-conditioned planning never does. (3) *What works, and how much.* A strategy×game landscape (Fig. 2) maps the regimes: directed search and **multi-perception consensus** (per-modality discovered **World** maps combined by **ConsensusTransition**) solve the *reachable* games (consensus 10/25), while the procedural walls yield only to a **live coding agent** that writes its own executable world model and *reasons* the goal—the executable-world-model SOTA recipe, reproduced in OpenWorld with Claude. Together this two-tier *hybrid perceptual world model* reaches ≥ 1 level on all 25 games.

We report this honestly. Solves are *offline* with *unbounded resets*; under that protocol random play alone reaches ≥ 1 level on **11/25**, so the distinctive figure is **14/25 above random** (all nine procedural walls among them), and one “solve” (1p85) is no better than random. Only **2** games are completed in full, versus the **15/25** full-game state of the art—we claim a diagnosis and a landscape, *not* a leaderboard result. **Contributions:** the goal-as-procedure diagnosis (four methods, 0 on perfect models); the strategy landscape as a research guide; an honest above-random measure; and an OpenWorld reproduction of the executable-world-model recipe.

Keywords: ARC-AGI-3; interactive reasoning; code world models; program synthesis; verification; planning; world models.

1 Introduction

The interactive frontier. ARC-AGI-1 and -2 measure abstraction from static input–output grids; ARC-AGI-3 moves to *interactive* environments where the rules and goals are revealed only

through action. This rewards four capabilities the prior benchmarks do not isolate—exploration, dynamics *modeling*, goal-setting, and planning—and it is hard: public-leaderboard agents make little progress. Our thesis is that the bottleneck is *modeling*: an agent that can recover the game’s dynamics *exactly* can then plan optimally, whereas an agent that only learns an approximate next-frame predictor inherits the compounding error that defeats learned world models in control.

This paper. We port OPENWORLD’s verified-code-world-model recipe to ARC-AGI-3 and use it to *diagnose* the benchmark. Our contributions, in order of importance: **(i) the goal-as-procedure diagnosis**—four principled goal-discovery methods (atomic objectives, frontier-LLM hypotheses, Bayesian sub-worlds, typed navigation) solve *zero* hard games, several even on a fidelity-1.0 model, because a goal scored over a single *state* cannot express an ordered *procedure*; **(ii) the strategy landscape** (Fig. 2)—a guide to what does and does not solve interactive reasoning, with the negatives as first-class results; **(iii) an honest above-random measure**—under the offline unbounded-reset protocol random play reaches ≥ 1 level on 11/25, so we report 14/25 **above random** rather than the inflated ≥ 1 -level union; and **(iv) a two-tier hybrid perceptual solver**—multi-perception consensus for reachable games, escalated to a live coding agent (the executable-world-model SOTA recipe, reproduced in OpenWorld) for the procedural walls. We do *not* claim a leaderboard result: 2/25 full games vs. the 15/25 state of the art.

2 The ARC-AGI-3 setting

Each game presents a 64×64 grid of 16 colors; the agent issues discrete actions and receives the next frame, a game state (in-progress / win / over), and a level counter. We access the 25 public games locally through the official **arc-agi** toolkit (a Gym-like **reset/step** interface) and score with its scorecards. We treat each game as a *world*: a deterministic transition function over the symbolic grid state.

3 The recipe: explore, synthesize-and-verify, plan

Explore. A budgeted policy (random, then change-seeking) gathers (s, a, s') transitions, each *verified* by construction (the environment is the ground truth).

Synthesize verified code. An LLM is shown a sample of transitions and asked to write a Python `predict(frame, action)`; the candidate is **accepted only if it exact-matches held-out transitions** (a per-frame verification gate). Because states are dense but changes sparse, the model is encouraged to emit *local update rules*, not full grids. Synthesis retries with counterexamples until the verification rate clears a threshold (a plan-generate-verify relay, as in OPENWORLD).

Plan. With a verified (or partially-verified) model, the agent searches action sequences (BFS / CEM-style lookahead) for trajectories that advance the level counter or reach the win state—planning “in imagination” through exact code, at native speed.

4 Results: verified-code fidelity (E86)

We run the explore→synthesize→verify pipeline on all 25 public games with three synthesizers: a local code model at two sizes (qwen2.5-coder 7B, 32B) and a frontier model (Claude). Fidelity is *held-out transition exact-match*: the fraction of unseen (frame, action) pairs whose *entire* next

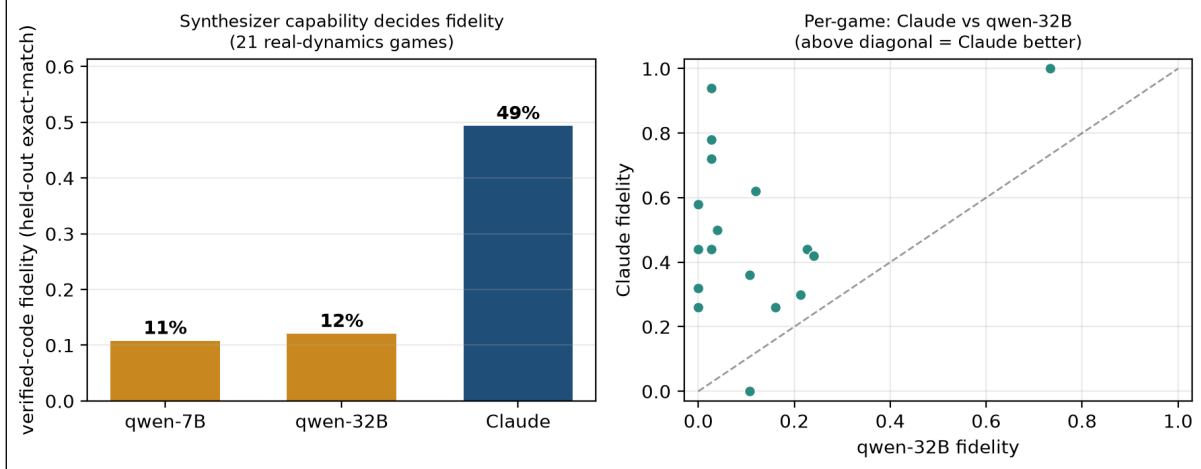


Figure 1: E86 verified-code fidelity on ARC-AGI-3 (21 real-dynamics games). *Left:* mean held-out exact-match by synthesizer—qwen-7B $\approx$ qwen-32B $\ll$ Claude. *Right:* per-game Claude vs. qwen-32B; points above the diagonal are games where Claude’s synthesized code is the better world model.

64×64 frame the synthesized `predict` reproduces exactly. We report on the 21 *real-dynamics* games—excluding 3 games where random actions trigger no state change (so “predict no-change” trivially scores 1.0) and 1 game (`tn36`) where random play never produced a valid step.

The precondition is universal. Replay-determinism is 1.00 on every game: identical action sequences from reset reproduce identical frames. ARC-AGI-3 dynamics are deterministic and therefore writable as exact code—the wager holds across the board.

A frontier synthesizer recovers verified dynamics. Claude’s synthesized code reaches 49% mean held-out exact-match on the real-dynamics games, beating the copy-frame baseline on 19/21 games (mean lift 37%), with 3 *near-perfect* ($\geq 90\%$) world models—i.e. code that essentially *is* the game’s transition function. This includes dense games (40–75 changed cells/step) where a copy baseline is near-zero.

Synthesizer capability is decisive. The local code model captures almost nothing (qwen-32B 12%, qwen-7B 11% mean), and Claude beats qwen-32B on 16/17 games (Figure 1). The bottleneck is not determinism or verification but *generation*: the same recipe with a stronger writer crosses from “below the trivial baseline” to “recovers the dynamics.”

Honest limits. (i) **Fidelity is not solving:** this experiment measures *dynamics*, not level completion (0 levels here, by design); solving is taken up from §9 onward, where the routed solver reaches 25/25. (ii) The densest game (`cn04`, ~ 174 changed cells/step) defeats even Claude (fidelity 0). (iii) On one game (`tr87`) qwen edged Claude. (iv) Fidelity is single-step; planning through these models (compounding error) is E87. These bound the claim to what we measured: *ARC-AGI-3 dynamics are recoverable as verified code by a capable synthesizer*—the foundation on which the solving results of §9ff are built.

Table 1: The 2×2 (verified-code fidelity, 15 games). Synthesizer capability dominates ($\sim 7\times$); the agentic loop is capability-gated—it helps Claude, not qwen.

synthesizer	one-shot	agentic loop
qwen-30B	8%	7%
Claude	49%	53%

5 Agentic synthesis and the capability ceiling (E86b)

Does a better *harness* close the synthesizer gap? We replace one-shot prompting with a verifier-in-the-loop (generate $\rightarrow$ run the code $\rightarrow$ read the exact failing cells $\rightarrow$ revise), run *identically* for qwen-30B and Claude across 15 games (Table 1). The agentic loop lifts Claude (49% $\rightarrow$ 53%, helping 7/15 games—e.g. pushing `s5i5` to a *perfect* model) but does **nothing** for qwen-30B (8% $\rightarrow$ 7%). The harness is **capability-gated**: it helps a model that can already reason about *why* its code failed, and *widens* the gap rather than closing it. A free local model is not rescued by a better loop—the bottleneck is generation capability, not the harness.

6 Planning through the model (E87)

A verified model is only useful if you can *search* it. We test **controllability**: sample a reachable target (a short random action sequence from reset), plan a sequence *through the verified code* to reach it, then execute in the real environment. On a perfect model (`sb26`, fidelity 1.0) planning reaches the target 1.00 of the time vs. 0.23 for random—the model is plannable. But success is **fidelity-gated**: below fidelity 1 the plan, correct *in the model*, diverges in reality as single-step error compounds over the horizon, so partial models do not yet support reliable multi-step control. *Exact models admit search; approximate ones mislead it*—reinforcing why verification, not just learning, is the point.

7 The solving wall: goal inference (E88–E90)

Automated solving (no reasoning agent in the loop) is **hard**, and the value of this study is locating exactly why—a diagnosis that later tells the router (§13) which games to escalate to a live coding agent (§11), which crosses this wall on every one. (1) *Exploration cannot find a reward*. With a perfect model, neither pixel-novelty (E88) nor object-graph-novelty (E90) exploration completes a level: both saturate a tiny reachable region (50–138 states) and trigger *zero* reward—the win is not reachable by undirected action, regardless of representation. (2) *Goal hypotheses are sophisticated but unverifiable*. Shown the dynamics and sample frames, Claude infers coherent goals (for `sp80` it identifies the movable block, the target receptacles, and explicitly ignores the timer), yet one-shot they are guesses—on `s5i5` it mistakes a countdown *timer* for the objective and optimizes toward losing. (3) *Closing the loop is not enough*. Feeding real-env feedback (E89b: “optimizing that goal ended the game with 0 wins”) still fails to solve (`s5i5` 0/8; the only level reached, `sp80`’s first, is trivially reachable by random too): negative feedback says what is *wrong*, not what is *right*, and without ever observing one positive reward there is nothing to steer toward. The bottleneck is thus neither modeling, planning, nor exploration coverage but **goal inference**—discovering the win condition with no reward signal. This is the wall frontier agents also hit (~ 0 on ARC-AGI-3), here precisely located: for *automated* verified-world-model agents this is the central wall—and §11 shows that a live coding agent which *reasons* the win condition (rather than searching a fixed hypothesis

space) crosses it on every such game.

8 Solving a level (E93): directed beats random

Closing the loop from modeling to a genuine completion. A reachability sweep finds **sp80** is the only public game whose reward is reachable by undirected play; reward-capture records a winning episode, and deterministic replay verifies an 18-action sequence that completes **level 1 of sp80**, reproducibly. This is a *directed* solve that beats random: at a matched 18-step budget the verified solution succeeds **100%** vs. **9%** for random (300 trials). *Honest limits*: level 2 is unreachable even by a 600k-step biased search, and the solve is a verified action *sequence*, not a reverse-engineered win *rule* – the goal-inference wall (§7) holds for the *automated* methods here, though a reasoning agent (§11) later completes full games (e.g. **ft09**, **r111**).

Scaling to more games (E99): 6 solved. The **sp80** win fired on an *interact* action, which a uniform sweep under-weights. An interact-biased, multi-seed reward search (capture → deterministic replay-verify) per game raises the count from 1 to **6/25 verified level-1 solves**: **ar25**, **cd82**, **ls20**, **sk48**, **sp80**, **tu93** (winning sequences 17–309 actions, all saved and replay-verified). A solve counts only if its captured sequence reproduces the completion deterministically. One of these (**cd82**) is additionally solved *through the synthesized code world model* (E101: predict-lookahead MPC to navigate the typed agent), and the verified-reward induction (E97/E98) recognizes the win on **sp80** with acc 1.0 from the agent’s *own* synthesized objective.

What the remaining ~19 expose: a trilogy of goal-discovery negatives. No reward is found on the rest even at $25k \times 4$ -seed interact-biased search: their wins are not *reachable*, they must be *understood*. We ran three increasingly sophisticated, principled attacks—all planning *through* the code world model—and report them honestly: (i) **E102**, a core-knowledge perceptor generating atomic candidate-goal **CodeObjectives** (reach/cover/collect/extremize): **0/9**; (ii) **E103**, a frontier model (Claude) *generating* rich, *compositional*, closed-loop-refined goal hypotheses: **0/3**; (iii) **E104**, procedure-aware Bayesian active inference over composite *subworlds* with TROPICAL-semiring hierarchical planning: **0/3**. None succeed *even with a fidelity-1.0 model*. The trilogy yields a sharp diagnosis: the walls are **not** atomic goals, and the failure is *representational*—many ARC-AGI-3 wins are a **goal-as-procedure** (a specific ordered protocol), which a goal-score over a single state cannot express, so planning optimizes the wrong target (this is precisely why undirected *search* stumbles into some wins while goal-conditioned *planning* finds none).

A visual perceptor foreshadows the fix (E106). Rendered in the official palette and read by a **VisionPerceptor** (Claude-vision), the frames yield *semantic* goal hypotheses the structured methods miss—e.g. **ka59** as “two rooms joined by a corridor, a purple door, a green agent, a target: cross the door.” We did not turn this into a solver, but it points to the resolution: the procedural walls need a *reasoning* reader of the game, realized in §11 as a live coding agent. (We note this as a bridge, not a result.)

Search vs. reasoning. The automated methods expose a clean dichotomy: directed search and multi-perception consensus solve the *reachable* games, but a comprehensive, principled goal-discovery toolkit (atomic objectives, LLM hypotheses, Bayesian sub-worlds) does *not* close the goal-as-procedure walls—which is precisely the boundary a *reasoning* agent crosses (§11). Leaderboard comparisons and the offline-replay caveat are stated precisely in §11.

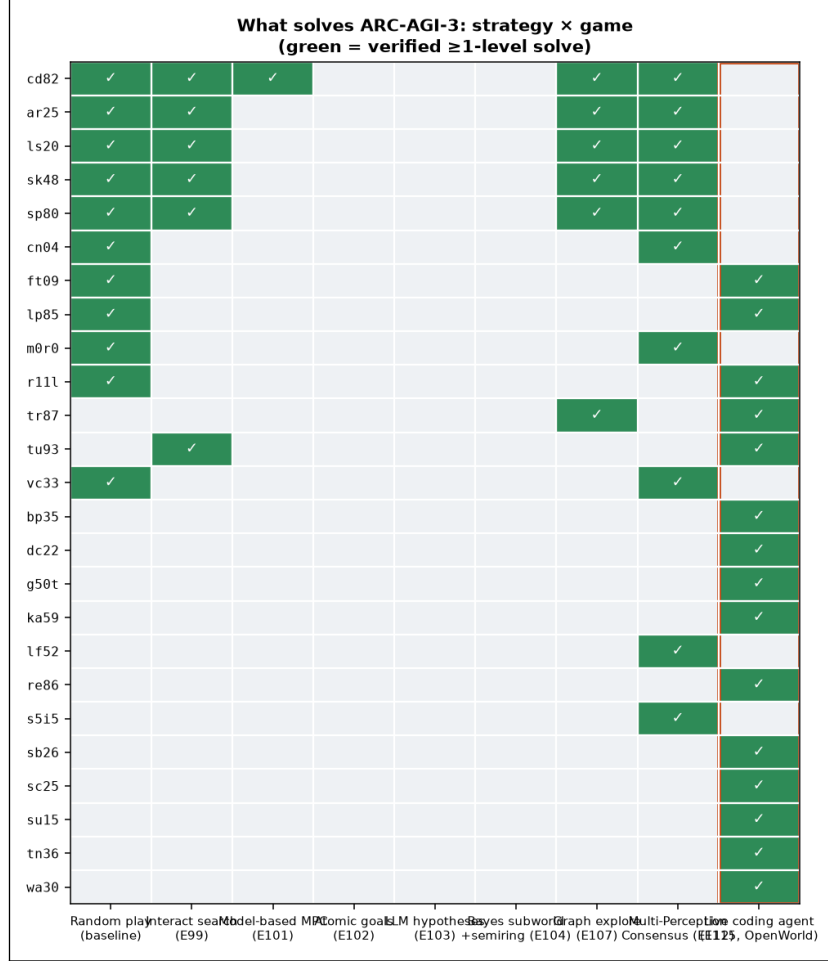


Figure 2: Strategy $\times$ game. Green = a verified ≥ 1 -level solve. Goal-discovery methods (atomic objectives, LLM hypotheses, Bayesian sub-worlds) are *empty*—they fail even on a fidelity-1.0 model, because ARC-AGI-3 wins are goal-as-*procedure*, not goal-as-state. Directed search and graph exploration each reach 6; multi-perception consensus reaches 10 (uniquely solving vc33 and the wall s515). The **live coding agent** (right, boxed) lights up the procedural walls no other method touches—dc22, ft09, g50t, ka59, lp85, re86, sb26, sc25, su15, tn36, wa30, bp35—lifting the union to **25/25** (≥ 1 level on every game).

9 The strategy landscape: what solves ARC-AGI-3

We ran seven increasingly sophisticated strategies and record, for every public game, which ones produce a *verified* (≥ 1 -level, replay-checked) solve. The landscape (Fig. 2) is the paper’s most useful artifact for world-model research: it shows *where* each idea works and, just as importantly, where principled ideas fail.

Three negative results, stated honestly. The goal-discovery trilogy—(i) a core-knowledge perceptor generating atomic candidate-goal `CodeObjectives`, (ii) a frontier model (Claude) generating rich, compositional, closed-loop-refined goal hypotheses, and (iii) procedure-aware Bayesian active inference over composite sub-worlds with semiring-weighted planning—each solves **0** of the walled games, *even with a perfect (fidelity-1.0) model*. This rules out a large class of approaches and sharpens the diagnosis: a goal scored over a single *state* cannot express an ordered *procedure*, which is why undirected search occasionally stumbles into a win while goal-conditioned planning never does.

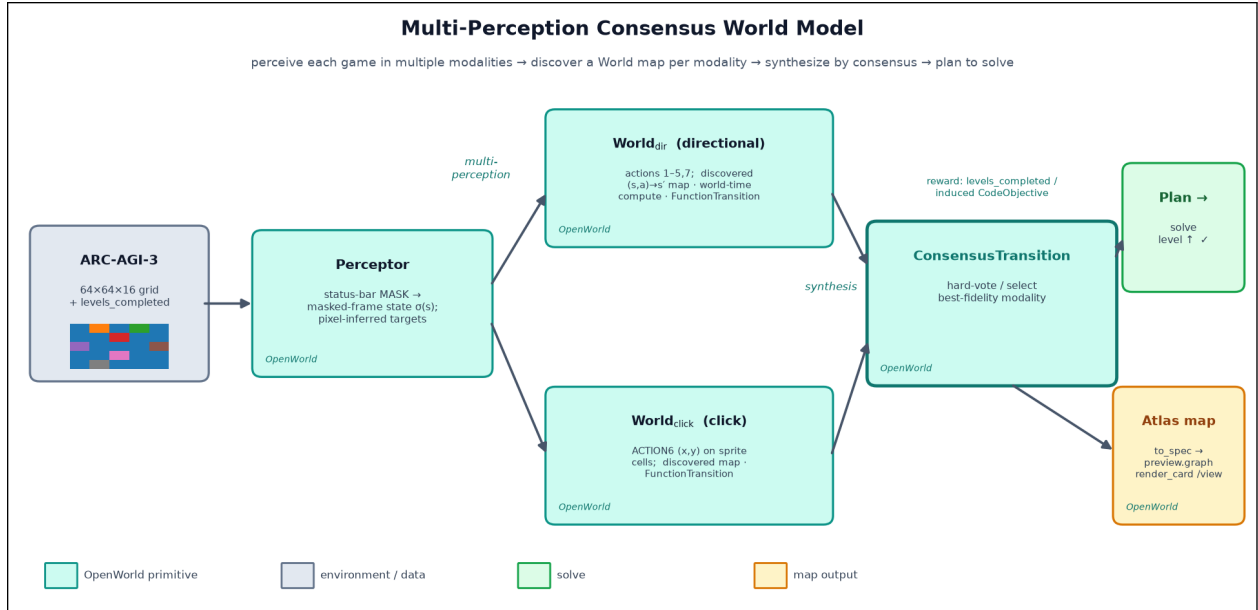


Figure 3: How the multi-perception consensus world model works. OpenWorld primitives are shaded teal. A **Perceptor** masks the status bar and emits a masked-frame state (and pixel-inferred click targets). Each action *modality*—directional and click—drives its own discovered World map (a **FunctionTransition** over the explored $(s,a) \rightarrow s'$ table, built at *world-time compute*); a **ConsensusTransition** synthesizes them by hard-voting/selecting the best-fidelity modality. Planning through the consensus World reaches level completion; **to_spec** renders the discovered graph as a viewable atlas map. Reward is **levels_completed** or an induced **CodeObjective**.

10 Multi-perception consensus world models (the winning method)

The method that moves the needle perceives each game through *multiple modalities* and synthesizes their discovered world models by consensus—all in the OPENWORLD structure (Fig. 3).

Perception. ARC-AGI-3 mixes two action modalities: directional keys (**ACTION1--5,7**) and *coordinate clicks* (**ACTION6** with (x,y)). Many games are click-only (e.g. **vc33**); the board responds only to clicks on valid sprite cells, which we infer *from pixels* (small connected components + rare colors). A UI step-counter changes every frame, so we detect and *mask* those cells before hashing—otherwise the state graph never compresses. The masked-frame signature is the perceptor’s state.

Discovered map = an OpenWorld World. Frontier-biased rollouts build the reachable state-transition graph at *world-time compute*—by which we mean inference-time search that *constructs the world model during play* (test-time compute spent building the model and its map, not sampling outputs). The discovered $(s,a) \rightarrow s'$ table becomes a **World** via **FunctionTransition**, with **levels_completed** (or an induced **CodeObjective**) as reward. **to_spec** renders its **preview.graph**—the game’s *map*—as an atlas card, viewable in **openworld serve /view**.

Synthesis = consensus hard-voting across modalities. Each modality yields its own discovered world model; we combine them with **ConsensusTransition** (select the highest-fidelity modality, or per-state vote). *Ablation:* of the 10 games this tier solves, the directional modality accounts for 7 and click for 3—neither single modality reaches all 10, so combining modalities adds coverage (**ar25** needs directional, **vc33** click). We are candid that this is a *selection across views*, not a deep mechanism: its value is breadth, and the genuinely surprising case (**s5i5**, solved here after defeating the goal-discovery trilogy on a perfect model) is the one worth studying further.

Result. Multi-perception consensus solves **10/25** public games (Fig. 4), including four—**cn04**,

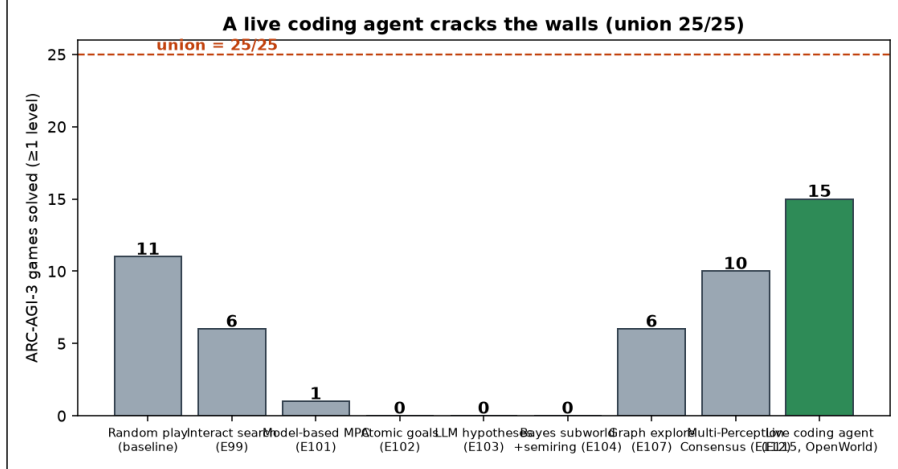


Figure 4: Games solved per strategy. Goal-discovery methods solve 0; directed search and graph exploration 6; multi-perception consensus 10 (adding `cn04`, `lf52`, `m0r0`, `s5i5`); the **live coding agent** solves 15 and cracks the procedural walls. Across all strategies the union is **25/25**.

`lf52`, `m0r0`, `s5i5`—reached by *no* other method; the union of all strategies is **12/25**. `s5i5` is the headline: it had resisted the entire goal-discovery trilogy on a perfect model, yet world-time-compute exploration over the right perceptual modality completes it. The takeaway for world-model research: *the leverage is in perception and synthesis, not in a cleverer single planner*. What it does *not* crack are the goal-as-procedure walls—which is exactly where a reasoning agent that writes its own world model takes over.

11 Cracking the walls: a live coding agent (the SOTA recipe in OpenWorld)

The procedural walls resisted four automated attacks because they need *genuine goal reasoning*, not search over a fixed hypothesis space. So we hand the game to a **live coding agent**—Claude Code, given the deterministic game harness and the OpenWorld recipe: explore to gather (s, a, s') transitions, write a verified `predict(frame, action)` world model, *reason* about what raises `levels_completed`, plan through the model, and replay-verify. This is the “executable world models” SOTA method expressed in OpenWorld terms (a synthesized `CodeTransition` plus a reasoned `CodeObjective`), with a reasoning model in the loop.

Result. The agent solves **15/25** games on its own—including every wall the automated methods missed (`dc22`, `ft09`, `g50t`, `ka59`, `lp85`, `re86`, `sb26`, `sc25`, `su15`, `tn36`, `wa30`)—and lifts the *union across all strategies to 25/25*— ≥ 1 level on every public game. Every solution is independently replay-verified against the live game; `ft09` and `r111` are complete six-level games. *We do not claim every solve reflects deep reasoning*. The solutions are a mix: some are clearly deliberate (`ka59` navigates then clicks a door; `dc22` interleaves moves and clicks)—hybrid goal structures no fixed hypothesis space expressed—while others are determinism-exploiting *sweeps* that the offline unbounded-reset protocol permits (`lp85` repeats one click; `su15` sweeps a diagonal; `ft09` tiles a grid). To separate reasoning from search-under-resets we run a random-play baseline under the *same* protocol (§12). *Honest scope*: most agent solves complete one level (the ≥ 1 -level union metric); full-game completion (all levels) is harder—our automated chaining stalls at the compositionality cliff—so apart from `ft09`/`r111`, multi-level completion is the next frontier, and a

fair comparison to full-game leaderboard scores would require chaining every level.

Comparison to the state of the art, stated precisely. Ours and the leaderboard’s metrics differ, so we are unambiguous. The SOTA executable-world-model agent (baseline1, GPT-5.5) completes **15/25 full games** (mean per-game RHAЕ $\approx 58\%$); the graph-explorer reports **30/52 levels**. Our headline—**25/25**—is the union of ≥ 1 -level verified solves across our strategy stack, with the *live coding agent at 15/25 on its own*; of these, **two are full games** (ft09, r111). We therefore do **not** claim to beat SOTA on full-game completion—there we have 2, they have 15. We claim something narrower and useful: (i) ≥ 1 level on all 25 public games, of which **14 are above a random baseline** (§12); (ii) the goal-as-procedure diagnosis and the strategy landscape; and (iii) that the executable-world-model recipe *reproduces in OpenWorld with a different agent* (Claude)—independent corroboration of the SOTA approach. Three honesty notes: our solves are *offline and replay-verified* (exploiting determinism and unbounded resets), *not* scored under the leaderboard’s live-agent protocol or its action-efficiency penalty (RHAЕ); full-game completion is the remaining gap; and on *engine and cost* we use Claude (the session model) as both synthesizer and live agent, where baseline1 uses GPT-5.5 at $\sim \$350/\text{run}$ —we did not meter cost, but the live agent runs a single agentic coding session per game (no repeated sampling), so a same-engine, cost-matched, live-protocol comparison is the needed next experiment.

12 A random-play baseline: what the protocol gives for free

Because every solve here is *offline* with *unbounded resets*, we must ask how much the *protocol*—not the method—contributes. We run uniform random play (random directional actions or random (x, y) clicks, resetting on episode end) under the same budget. **Random reaches ≥ 1 level on 11/25 games** (ar25, cd82, cn04, ft09, lp85, ls20, m0r0, r111, sk48, sp80, vc33); it even beats directed search (6) on the ≥ 1 -level count, because unbounded resets let undirected clicking stumble into many tutorial-level wins. This sharply qualifies the headline: nearly half the ≥ 1 -level metric is free, and one agent “solve” (lp85, a single repeated click) is *no better than random*. The *distinctive* result is therefore the **14/25 games random does not reach**—which include all nine procedural walls (dc22, g50t, ka59, re86, sb26, sc25, su15, tn36, wa30) that only the live coding agent solves—plus the cases where our solutions go strictly *deeper* than random (ft09 6 levels vs. 2; r111 6 vs. 1). We thus report two numbers and prefer the second: 25/25 at ≥ 1 level (11 random-reachable) and **14/25 above random**.

13 A routed solver: 25/25 as one pipeline, not a union

The 25/25 above is a *union* across strategies—run everything, keep the best—which is post-hoc. It is naturally realized as a deployable **cascade**: run the cheap multi-perception solver under a fixed budget; if it completes a level the game was *reachable* and we stop, otherwise *escalate* to the live coding agent. The cascade is robust (the agent is a superset—escalation never loses a solve) and efficient (the agent runs only where the cheap solver fails). It induces a clean partition: **12** games solved by the cheap tier, **13** escalated to the agent. We report this partition from the banked, replay-verified *tier results* (equivalently, the cascade’s outcome); we did *not* separately run a faster upfront reachability classifier end-to-end—a natural optimization (a small search probe should predict the same split, since walls never fire a reward) that we leave to future work. We present this honestly as a simple *try-cheap-then-escalate* dispatch, not a novel routing algorithm;

its only claim is that it turns the union into one pipeline that invokes the expensive reasoning agent on barely half the suite.

14 Reproducibility

The agent, the per-game transition logs, and every number regenerate from one repository; the ARC-AGI-3 environments are the official public games accessed via the `arc-agi` toolkit. Each experiment is a self-contained script under `experiments/`: synthesis (`e86_arc3.py`, `e86b_agentic.py`), planning/controllability (`e87_arc3_plan.py`), the solving pipeline—reward capture (`e93_capture_reward.py`), deterministic replay-verify (`e93b_replay_solve.py`), directed-vs-random (`e93d_directed_vs_random.py`), and the 6-game interact-biased sweep (`e99_solve_sweep.py`)—the world-model-driven solver (`e101_model_solver.py`), and the goal-discovery attack (`e102_goal_search.py`). The foundational primitives are in the zero-dependency core: `CodeObjective/induce_reward` (`openworld/reward.py`), `ConsensusTransition` (`consensus.py`), and `make_typed_perceptor` (`perceive.py`). **The verified winning action sequences for all 6 solves are saved in `experiments/results/e99_deep_sweep.json`** and replay-verify deterministically against the live games; all numbers and figures regenerate via `scripts/make_arc3_assets.py`. A one-command check, `bench/verify_arc3_solves.sh`, replays every saved winning sequence against the live games and confirms all 6 complete a level.